

連立② - 重解をもつとき -

$$\begin{cases} a_{n+1} = p a_n + q b_n \\ b_{n+1} = r a_n + s b_n \end{cases}$$

$$a_{n+1} - \alpha b_{n+1} = \beta (a_n - \alpha b_n)$$

と変形して④等比型へ

Ⅲ B

$$\begin{cases} a_{n+1} = a_n - b_n \\ b_{n+1} = a_n + 3b_n \end{cases} \quad a_1 = 1, b_1 = 1$$

$$a_{n+1} - d b_{n+1} = \beta (a_n - d b_n) \dots \textcircled{1}$$

とあくと

$$a_n - b_n - d(a_n + 3b_n) = \beta (a_n - d b_n)$$

$$(1-d)a_n + (-1-3d)b_n = \beta a_n - d\beta b_n$$

よって

$$\begin{cases} 1-d = \beta \\ -1-3d = -d\beta \end{cases}$$

$$\therefore (d, \beta) = (-1, 2)$$

これを①に代入して

$$a_{n+1} + b_{n+1} = 2(a_n + b_n)$$

ゆえに

$$\{a_n + b_n\} \text{ は } \textcircled{1} a_1 + b_1 \textcircled{2} 2 \text{ の等比数列}$$

$$a_n + b_n = (a_1 + b_1) \cdot 2^{n-1}$$

$$= 2^n$$

$$\text{これと } a_{n+1} = a_n - b_n \text{ より}$$

$$a_{n+1} = 2a_n - 2^n \quad \leftarrow \text{Ⅲ B 指数型② に帰着}$$

この両辺を 2^{n+1} でわって

$$\frac{a_{n+1}}{2^{n+1}} = \frac{a_n}{2^n} - \frac{1}{2}$$

$$\frac{a_n}{2^n} = C_n \text{ とおくと } \leftarrow \frac{a_{n+1}}{2^{n+1}} = C_{n+1}$$

$$\begin{cases} C_{n+1} = C_n - \frac{1}{2} \\ C_1 = \frac{a_1}{2^1} = \frac{1}{2} \end{cases} \quad \leftarrow \text{Ⅳ 等差型に帰着}$$

よって

$$C_n = \frac{1}{2} + (n-1) \cdot \left(-\frac{1}{2}\right)$$

$$= \frac{1}{2}(2-n)$$

$$\text{よって, } \frac{a_n}{2^n} = C_n \text{ より}$$

$$a_n = 2^n \cdot C_n$$

$$= 2^n \cdot \frac{1}{2}(2-n)$$

$$= 2^{n-1}(2-n) //$$

$$\text{また, } a_n + b_n = 2^n \text{ より}$$

$$b_n = 2^n - a_n$$

$$= 2^n - 2^{n-1}(2-n)$$

$$= 2^{n-1}\{2 - (2-n)\}$$

$$= n \cdot 2^{n-1}$$